

LESSON 2-7 • SESSION 2 • PRACTICE • FORM B

Dividing by a Decimal

Learning target – I can divide by a decimal by making the divisor a whole number first. • 6.NS.B.3

REMEMBER

- Make the divisor whole: multiply **both** numbers by the **same** power of 10.
- Count the divisor's decimal places. One place: $\times 10$. Two places: $\times 100$. Example: $5 \div 0.25 = 500 \div 25 = 20$.
- Dividing by a number less than 1 gives a quotient **greater** than the dividend.

WORKED EXAMPLE

How many 0.7-liter bottles can be filled from 8.4 liters?

Step 1 Write $8.4 \div 0.7$.**Step 2** Multiply both by 10: $0.7 \times 10 = 7$ and $8.4 \times 10 = 84$.**Step 3** $84 \div 7 = 12$. Check: $12 \times 0.7 = 8.4$.**Answer** 12 bottles**WORDS** **divisor** (*divisor*) • **equivalent division** (*división equivalente*) • **power of ten** (*potencia de diez*)

PRACTICE

show your work • label units

1. Rewrite with a whole-number divisor. Then divide.

a. $2.4 \div 0.15 =$ _____ b. $9.75 \div 0.25 =$ _____ c. $1.92 \div 0.08 =$ _____
d. $22.5 \div 1.5 =$ _____ e. $0.54 \div 0.09 =$ _____ f. $8.8 \div 0.4 =$ _____

- 2.
- Greenhouse Pours.**
- A greenhouse pours 8.4 liters of water into 0.7-liter pots. Later it pours 5.6 liters into 0.8-liter trays.

- a. Write the whole-number problem. How many pots? _____
b. Write the whole-number problem. How many trays? _____

Show your work.

- 3.
- Paint Cans.**
- A painter fills 1.1-liter cans from 14.3 liters of paint. How many cans are filled?

Hint Multiply both numbers by 10.

Show your work.

Answer _____ cans

- 4.
- One-Sided Slide.**
- For
- $7.2 \div 0.9$
- , Theo divided 7.2 by 9. What did his shortcut do to the quotient? Find the correct answer.

Hint Did Theo multiply both numbers by 10?

Show your work.

Quotient _____

- 5.
- Bagging Rice.**
- A store packs 30.6 kg of rice into bags that hold 1.8 kg each. How many bags?

Show your work.

Answer _____ bags

- 6.
- Kite Tails.**
- A spool holds 11.2 m of string. Each kite tail uses 0.8 m. How many tails?

Show your work.

Answer _____ tails

7. **Too Few Slides.** For $4.2 \div 0.12$, Mei moved the divisor's point two places but the dividend's point one place: $42 \div 12$. Explain her mistake and find the correct quotient.

Hint 0.12 has two decimal places, so multiply both by 100.

Show your work.

Quotient _____

8. **Paint Coverage.** A painter has 16.8 liters of paint. Each door takes 0.8 liter.

Hint Bigger amount per door means fewer doors.

- a. How many doors can the painter cover? _____ doors
- b. If each door took 1.2 liters, how many doors? _____ doors
- c. Why does the number of doors drop? _____

Show your work.

FIND THE ERROR

explain the mistake, then fix it

9. **Malik's Work.** For $4.8 \div 0.6$, Malik correctly rewrote the problem as $48 \div 6$, then wrote his answer as 0.8. Explain the mistake and give the correct quotient.

TEST PRACTICE

choose the best answer

10. Find $4.5 \div 0.5$.
- A 0.9

B 2.25

C 90

D 9
11. Multiplying the divisor and the dividend by 10 makes a division that is –
- A 10 times larger

B 10 times smaller

C equivalent

D impossible to solve

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

35 O	12 D	0.8 Q	17 I	13 V	1.3 X	7 I	14 S	21 R	3.5 Z
#2a		#2b	#3	#5	#6	#7	#8a		

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #____